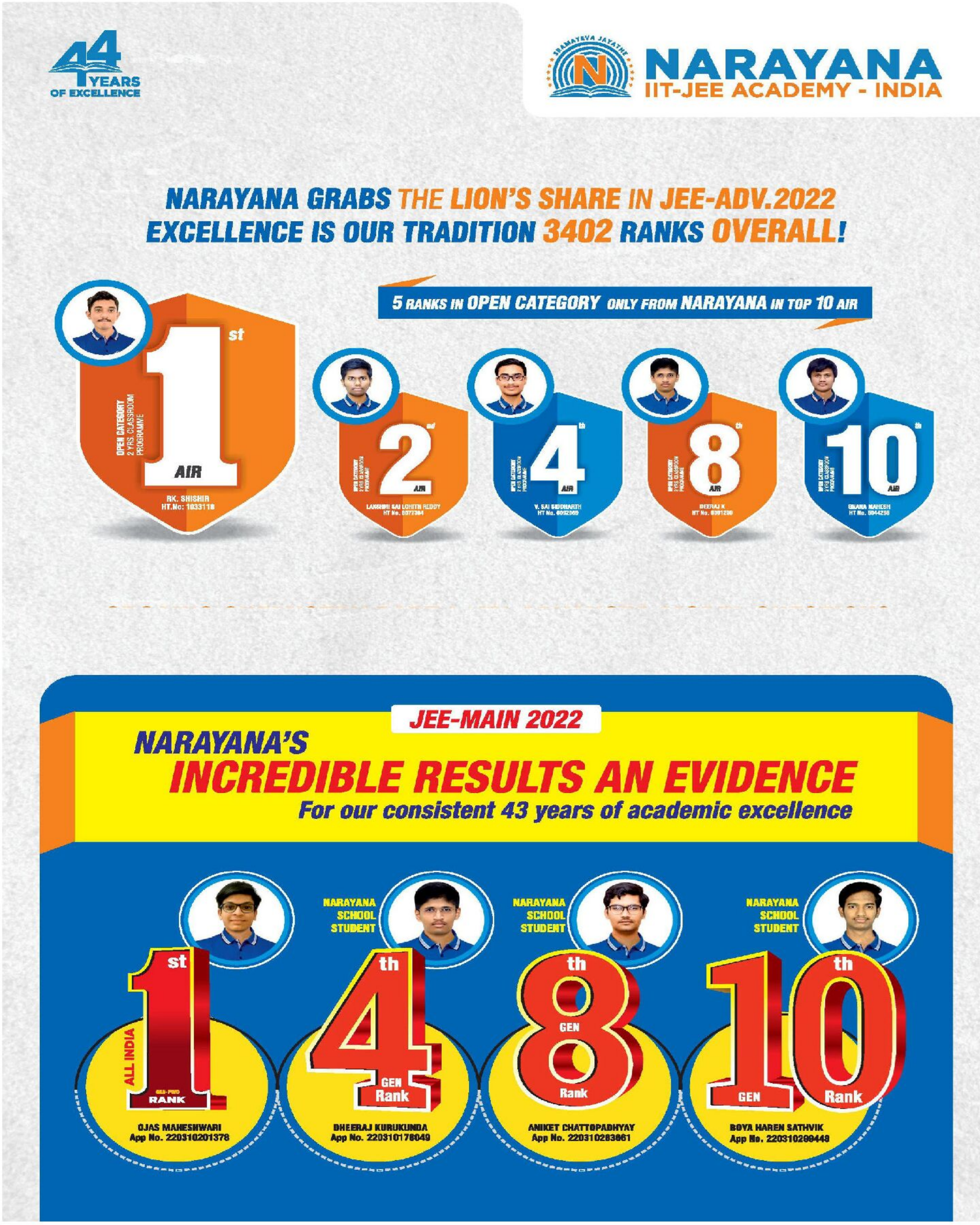


MATHS

ALGEBRA

JEE ADVANCED IMPORTANT QUESTIONS

ALGEBRA

MULTI ANSWER TYPE QUESTIONS

1. For all permissible values of x, consider $y = \frac{\sin 3x(\cos 6x + \cos 4x)}{\sin x(\cos 8x + \cos 2x)}$ and range of y is $(-\infty, a) \cup (b, \infty)$. If 2b is the first term of G.P and 'a' is its common ratio, then

$(S_8 \textit{ denotes the some of infinite terms of } G.P)$

(A) $b - a = \frac{10}{3}$ (B) $3a + b = 4$ (C) $s_\infty = 9$ (D) $s_\infty = \frac{27}{10}(a+b)$

2. It is given that the sequence $\{a_n\}$ satisfies $a_1 = 0,\ a_{n+1} = a_n + 1 + 2\sqrt{1 + a_n}$ for $n \in N$. Then

(A) $a_{100} = 9999$ (B) $a_{2001} = 4004000$ (C) $a_{2001} = 4002000$ (D) $a_{19} = 360$

3. Let $\{a_n\}$ consists of positive numbers and for any positive integer n, $\frac{a_n + 2}{2} = \sqrt{2s_n}$, where

$$s_n = \sum_{i=l}^{n} a_i.$$

(A) $a_{21} = 82$ (B) $a_{12} = 48$ (C) $a_{13} = 50$ (D) $a_{14} = 54$

4. If x,y,z are three distinct positive real numbers and are in H.P., Then $\frac{3x+2y}{2x-y} + \frac{3z+2y}{2z-y}$ is greater than

(A) 9 (B) 10 (C) 12 (D) 15

5. The sequence $\{a_n\}$, $n \in N$ satisfies $a_1 = 1$ and $5^{a_{n+1} - a_n} = 1 + \frac{1}{n + \frac{2}{3}}$. Then

(A) $[a_{501}] = 3$ (B) $[a_{207}] = 3$ (C) $[a_{223}] = 4$ (D) $[a_{625}] = 4$

(where [.] denotes greatest integer function)

6. If $n \in N$, n>5 then which of the following holds true ?

(A) $n^n > 1.3.5...(2n-1)$ (B) $2^n > 1 + n\sqrt{2^{n-1}}$

C) $\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + ...\frac{1}{2n} > \frac{1}{2}$ (D) $2^n < 1 + n\sqrt{2^{n-1}}$

7. Let $\{a_n\}$ be a sequence of real numbers such that $a_1 = 2, a_{n+1} = a_n^2 - a_n + 1$ for n = 2,3,…. . Let

$S=\frac{1}{a_1}+\frac{1}{a_2}+...+\frac{1}{a_{2018}},$

(A) $S<1-\frac{1}{(2018)^{2018}}$ (B) $S>1-\frac{1}{(2018)^{2018}}$ (C) $S<1$ (D) $S<1-\frac{1}{(2017)^{2017}}$

8. Let $a_k=\frac{k}{(k-1)^{4/3}+k^{4/3}+(k+1)^{4/3}}$ and $S_n=\sum_{k=1}^{n}a_k$, then

(A) $S_{26}>\frac{17}{4}$ (B) $S_{26}<\frac{17}{4}$ (C) $S_{999}<50$ (D) $S_{999}>50$

9. Let $S=\frac{1}{\sqrt{1}+\sqrt{3}}+\frac{1}{\sqrt{5}+\sqrt{7}}+\frac{1}{\sqrt{9}+\sqrt{11}}+...\frac{1}{\sqrt{9997}+\sqrt{9999}}$, then

(A) $S<24$ (B) $S>24$ (C) $S>18$ (D) $S<18$

10. When we divide the ninth term of an arithmetic progression by its second term get 5 as quotient and when we divide the thirteenth term of that progression by sixth term, we get 2 as a quotient and 5 as a remainder, then

(A) First term is 3 (B) Seventh term is 27 (C) First term is 4 (D) Third term is 9

11. The sequence of all positive palindromes are written in ascending order 1,2,3,4,5,6,7,8,9,11, 22,………… then

(A) The 2018^{th} positive palindrome is 1019101 (B) The 1998^{th} positive palindrome is 1001001

(C) The rank of 2140412 is 3139 (D) The rank of 1999999 is 2998

12. Given '2n' different objects arranged around a circle, the number of ways of choosing k of them so that no two of them are consecutive is equal to $(0\le k\le n)$

(A) ${}^{2n-k+1}C_k-{}^{2n-k-1}C_{k-2}$ (B) $\frac{2n}{k}\,{}^{2n-k-1}C_{k-1}$ (C) $\frac{2n}{(2n-k)}\,{}^{2n-k}C_k$ (D) $\frac{n(4n-4k+1)}{(2n-2k+1)}\,{}^{2n-k}C_k$

13. The number of ordered triplets (x,y,z) of non-negative integers satisfying the conditions.

$x+y+z\le 100$ and $x\le y\le z$

(A) If x is odd is 14724 (B) If x is odd is 14722 (C) If x is even is 16065 (D) If x is even is 16164

14. Let x be the number of 6 digit numbers, the some of whose digits is even and y be the number of 6 digit numbers, the some of whose digits is odd, then

(A) $x+y=9\times 10^5$ (B) $x<y$ (C) $x=y$ (D) $y=450000$

15. There are n lines in a plane, no two of which are parallel and no three of them are concurrent. Let the plane be divided by n lines in a_n parts, then

(A) $a_n=a_{n-1}+(n-1)$ (B) $a_n=a_{n-1}+n$ (C) $a_6=22$ (D) $a_{10}=56$

16. If $(1+x+2x^2)^{20}=a_0+a_1x+a_2x^2+......+a_{39}x^{39}+a_{40}x^{40}$ and $N=a_0+a_2+a_4+......+a_{38}$, then N is

divisible by

(A) 2^{20} (B) 2^{19} (C) 3 (D) 25

17. If $(1+x+x^2)^n = a_0 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, then the value of $a_1 - a_3 + a_5 - a_7 + ...$ is equal to

(A) $1\ if\ n = 4k$ (B) $-1\ if\ n = 4k+1$ (C) $0\ if\ n = 4k+2$ (D) $-1\ if\ n = 4k+3$

18. If the unit digit of $13^n + 7^n - 3^n, n \in N$, is 3 then possible value(s) of n is/are

(A) 27 (B)103 (C)11 (D) 101

19. Let $n \in N$, $n > 3$ then $2^{4n} - 2^n(7n+1)$ is divisible by

(A) 125 (B)196 (C)16 (D)49

20. Let $S_n = \sum_{k=0}^{n} \frac{{}^{n+k}C_k}{2^k}$, then

(A) $S_{16} = 2S_{17}$ (B) $S_{18} = 2S_{17}$ (C) $S_{12} = 4096$ (D) $S_{14} = \frac{1}{2}S_{15}$

21. If the coefficients of x^t and x^{t+1} in $S_n = \sum_{r=0}^{n}(1+x)^r$ where $t < n-2$ are equal, then

(A) n is odd (B) n is even

C) The sum of coefficients of $x^t\ and\ x^{t+1} = {}^{n+1}C_{t+2}$ (D) The sum of coefficients of $x^t\ and\ x^{t+1} = {}^{n+2}C_{t+2}$

22. Which of the following statement(s) is/are Incorrect.

(A) If A is any non-singular matrix, then $(tr(A)A)^{-1}$ always exists.

(B) If A is any square matrix of order 'n' such that $A = A^{-1}$, then $|A^2 + A^4 + A^6 + ... + A^{50}| = 625$ holds good for some n, $n \in N$

(C) If A is any non-singular matrix of order 2, then $|adj(|A|A)| = |A|^3$

(D) If A is any non-singular matrix of order 2, then $|adj(|A|A)| = |A|^2$

23. Let A and B are two square matrices such that AB=A, BA=B, then $(A^2 - AB + B^2)$ is always equal to

(A) A (B) B (C) A^2 (D) B^2

24. Let $A = \left[a_{ij}\right]_{3\times3}$ be a matrix such that $a_{ij} = \frac{|i-j|}{2}$, then

(A) A is a symmetric matrix (B) $2A^3 - 3A = I$ (C) $2A^3 - 3A + I = 0$ (D) A is not an orthogonal matrix

25. If A and B are square matrices such that AB=B and BA=A, then

(A) A is an idempotent matrix (B) $A^3+B^2=A^2+B^3$

(C) $A^3+B^3=A+B$ (D) A is an involuntary matrix

26. Let $A=\begin{bmatrix}1&0&0\\2&1&0\\4&2&1\end{bmatrix}$ and $B=[b_{ij}]$ is a 3x3 matrix such that $A^{100}-B=I$, then

(A $\frac{b_{31}+b_{32}}{b_{21}}=102$ (B) $b_{32}=b_{21}$

(C) B is lower triangular matrix (D) B is upper triangular matrix

27. Let 'Z' satisfy the equation $iZ^3+Z^2-Z+i=0$, then which of the following may be correct?

(A) $\arg(Z)=\frac{\pi}{4}$ (B) $Z=\frac{1}{\sqrt{2}}(-1+i)$ (C) $\arg(Z)=-\frac{\pi}{4}$ (D) $|Z|=1$

28. Let two distinct points $P(z_1)$ and $Q(z_2)$ lie on curve $C: z+\overline{z}=2|z-1|$ such that $\arg(z_1-z_2)=\frac{\pi}{4}$, then

(A) P,Q lie on ellipse (B) $\text{Im}(z_1+z_2)=2$

(C)P,Q lie on parabola

(D)Point of intersection of tangents to 'C' at P and Q lie on curve $\text{Im}(z)=1$

29. Consider a curve 'C' given by equation $|z-3-2i|=\left|z\cos\left(\frac{\pi}{4}-\arg z\right)\right|$, then

(A) represents a parabola (B) 'C' represents a ellipse

(C) Point $P\left(\frac{7+3i}{2}\right)$ lies on 'C' (D) Point $P\left(\frac{7+3i}{4}\right)$ lies on 'C'

30. If $1,z_1,z_2,z_3,......z_{n-1}$ are 'n' n^{th} roots of unity, then the value of $\frac{1}{3-z_1}+\frac{1}{3-z_1}+\frac{1}{3-z_1}+......+\frac{1}{3-z_{n-1}}$ is equal to

(A) $\frac{\sum_{r=1}^{n} r\,3^{r-1}}{\sum_{r=1}^{n} 3^{r-1}}$ (B) $\frac{\sum_{r=1}^{n-1} r\,3^{r-1}}{\sum_{r=1}^{n} 3^{r-1}}$ (C) $\frac{n3^{n-1}}{3^{n-1}}-\frac{1}{2}$ (D) $\frac{n3^{n}}{3^{n-1}}-\frac{1}{2}$

31. The locus of point $P(z)$ such that z, z^2 and z^3 are the vertices of a right angled triangle can be

(A) line, $\text{Re}(z)=-1, z\neq -1$ (B) line, $\text{Re}(z)=0, z\neq 0$

(C) Circle with centre (-1) and radius $1, z\neq 0$

(D) Circle with centre $\left(-\frac{1}{2}\right)$ and radius $=\frac{1}{2}, z\neq -1,0$

32. If the equation of $x^2+(p+ip')x+(q+iq')=0$ has two equal roots, then $(p, p', q, q'\ are\ real)$

(A) $p^2+(p')^2=4q$ (B) $pp'=4q'$ (C) $p^2-(p')^2=4q$ (D) $pp'=2q'$

33. Let $A(z_1), B(z_2), C(z_3)$ are points in complex plane such that $z_1|z_2-z_3|-z_2|z_3-z_1|-z_3|z_1-z_2|=0$, then which of the following nay be correct?

(A) A, B, C are collinear such that A lies between B and C

(B) A, B, C are collinear such that B lies between A and C

(C)A, B, C are collinear such that C lies between A and B

(D) O(0) is the centre of circle which touches the sides of triangle ABC.

34. Let z_1, z_2, z_3 be three distinct non zero complex numbers, then

(A) There always exist real numbers p, q, r such that $pz_1+qz_2+rz_3=0$

(B) If $pz_1+qz_2+rz_3=0$ and $p+q+r=0, p, q, r \in R,$ then z_1, z_2, z_3 are collinear.

(C)There always exist complex numbers p, q, r such that $pz_1+qz_2+rz_3=0$ and $p+q+r=0$

(D) There always exist real numbers p, q, r such that $pz_1+qz_2+rz_3=0$ and $p+q+r=0$.

35. Consider the equation $E: 9^{-|x-2|}-4.3^{-|x-2|}-a=0, a \in R$, and S be the set of all real values of 'a' for which eqn. (E) has solutions. Then value of X satisfying (E) is/are

(A) $x=2+\log_3(2-\sqrt{4+a}), a \in S$ (B) $x=2-\log_3(2-\sqrt{4+a}), a \in S$

(C) $x=2+\log_3(2+\sqrt{4+a}), a \in S$ (D) $x=2-\log_3(2+\sqrt{4+a}), a \in S$

36. Let x, y are positive real numbers and satisfy the system of equations $x^{x+y}=y^n$ & $y^{x+y}=x^{2n}y^n$ where $n>0$, then

(A) $x=\dfrac{2n+2-\sqrt{4n+1}}{2}$ (B) $x=\dfrac{\sqrt{1+8n}-1}{2}$

(C) $y=\dfrac{\sqrt{4n+1}-1}{2}$ (D) $y=\dfrac{4n+1-\sqrt{1+8n}}{2}$

37. For each ordered pair of real numbers (x, y) satisfying $\log_2(2x+y)=\log_4(x^2+xy+7y^2)$.

There is a real number k such that $\log_3(3x+y)=\log_9(3x^2+4xy+ky^2)$.

Then possible value of k is/are:

(A) 9 (B) 12 (C) 14 (D) 21

38. Let a > 1, b > 1, satisfy $\log_a(\log_a(\log_a 2)+\log_a 24-128)=128$ and $\log_a(\log_a b)=256$, then

(A) $a^2=(128)^{\frac{1}{128}}$ (B) $b = 2^{194}$ (C) $b = 2^{192}$ (D) $a^2=(64)^{\frac{1}{64}}$

39. The possible real values of k for which the equation $\log_{10}(kx)=2\log_{10}(x+2)$ has exactly one solution is/are:

(A) -5 (B) -4 (C) 4 (D) 8

INTEGER TYPE QUESTION

40. The number of integral value of 'a' for which real value of x that satisfies the inequality $ax^2+\left(1-a^2x\right)-a>0$ does not exceed two in absolute value.

41. Let $P(x)$ be a polynomial with real coefficients and leading coefficient unity satisfy the identity $(x-8)P(2x)=8(x-1)P(x)$. Then $\left[\sqrt{P(10)}\right]$ is equal to ([.] denotes greatest integer function).

42. Let $P(x)=0$ be a fifth degree polynomial equation with integer coefficients that has a integral root 'α'. If $P(2)=13$ and $P(10)=5$ then $\left[\frac{a}{2}\right]$ is equal to ([.] denotes greatest integer function).

43. The number of integral values of 'a' for which the equation $\cos 2x + a\sin x = 2a-7$ possesses solution is

44. The number pf ordered pairs $(x,y), x,y\in R$ satisfying equation $4^{\left|x^2-8x+12\right|-\log_4 7}=7^{2y-1}$ and the inequality $|y-3|-3|y|-2(y+1)^2\ge 1$ is

45. The number of integral values of n for which the equation $nx^2+(n+1)x+(n+2)=0$ has rational roots only is equal to

46. If p_1, p_2 are the roots of the quadratic equation $ax^2+bx+c=0$ and q_1, q_2 are the roots of the quadratic equation $cx^2+bx+a=0$ such that p_1, q_1, p_2, q_2 is an A.P. of distinct terms, then $a+c$ equal to $(a,b,c\in R)$.

47. The number of integral values of 'a' for which the equation $\left(x^2+x+2\right)^2-(a-3)\left(x^2+x+2\right)\left(x^2+x+1\right)+(a-4)\left(x^2+x+1\right)^2=0$ has atleast one real root is

48. Let $A_1A_2A_3......A_{200}$ be a regular 200-sided convex polygon. Make the diagonals $A_iA_{i+9}, i=1,2,3,....,200,$ where $A_{i+200}=A_i$ for $1\le i\le 9$. Let N be the number of distinct points of intersection formed inside the polygon by these 200 diagonals, then the sum of the digits of N is

49. There are given points on a circle and each two points are connected by a segment. Suppose that any three segments are not concurrent, so any three intersecting segments form a triangle inside the circle. Let N be the number of triangles formed, then sum of digits of N is

50. All digit numbers containing each of the digits 1,2,3,4,5,6,7 exactly one and not divisible by 5, are arranged in the increasing order. Let N be the 2018th number in this list, then the last digit of N is

51. Determine the number of ordered pairs (a,b) of integers such that $\log_a b + 6\log_b a = 5,\ 2 \le a \le 2019$ and $2 \le b \le 2019$

52. Let N be the number of all 5 digit numbers each of which contains the block 15 and is divisible by 15. Then the last digit of N is

53. Let N be the number of 6 digit numbers such that the digits of each number are all from the set $\{1,2,3,4,5\}$ and any digit that appears in the number appears atleast twice. Then the last digit of N is

54. Let N be the number of all 6 digit numbers such that the sum of their digits is 10 and each of the digits 0, 1, 2, 3 occurs atleast once in them, then the last digit of N is

55. 200 moviegoers queen up for tickets at the box office. The price admission is Rs. 100 and the cashier has no change initially. Of the moviegoers, 100 have rupee 100 note only and 100 have rupee 200 note only. The probability that all of the moviegoers can collect their tickets without facing any problem of change is $\frac{1}{p}$, then p is equal to

56. A bag has 10 balls, 6 balls are drawn in an attempt and replaced. Then another draw of 5 balls is made from the bag, the probability that exactly two balls are common to both the drawn is $\frac{m}{n}$, where m,n are coprime natural numbers, then n is equal to

57. 32 players ranked 1 to 32 are playing a knochout tournament. Assume that in every match between two players, the better ranked player wins. The probability that ranked 1 and ranked 2 players are winner and runner up respectively is $\frac{m}{n}$ where m,n are relatively prime natural numbers, then $m+n$ is equal to

58. The sequence $a_1, a_2, a_3, ...$ is a G.P. of positive numbers. Given that $\log_8 a_1 + \log_8 a_2 + \log_8 a_3 + ... + \log_8 a_{12} = 2006,$ find the number of distinct value of a_1.

59. Find b, $b \ge 2$ satisfying the equations $3\log_b\left(\sqrt{x}\log_b x\right) = 56$ and $3\log_{\log_b x}(x) = 54,$ where $x > 1$.

Comprehension type questions

Passage – 1 (Q. No. 60 to 62)

Let $\{a_n\}, a_n 0, n \in N, a_1 = 1$ be a sequence of real numbers and $S_n = a_1 + a_2 + a_3 + ... + a_n$. Also it is given that $S_n = \frac{1}{2}\left(a_n + \frac{1}{a_n}\right)$

60. $S_{100} + S_{400} + S_{900}$ is equal to

A) 10 B) 20 C) 30 D) 60

61. $\lim_{n\to\infty}\left(\sqrt{n}\, a_n\right)$ is equal to

A) 1 B) $\frac{1}{2}$ C) $\frac{1}{\sqrt{2}}$ D) 0

62. $\left[\sum_{k=1}^{100} \frac{1}{S_k}\right]$ is equal to ([.] = denotes greatest integer function)

A) 18 C) 19 C) 20 D) 17

Passage – 2 (Q.No. 63 to 65)

If $\phi(r) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + ... + \frac{1}{r}$ and $\sum_{r=1}^{n}(2r+1)\phi(r) = P(n)\phi(n+1) - Q(n)$, where $P(n)$ and $Q(n)$ are polynomial functions of 'n', then

63. $\sum_{r=0}^{10} P(r)$ is equal to

A) 235 B) 506 C) 285 D) 385

64. $\sum_{r=0}^{\infty} \frac{1}{Q(r)}$ is equal to

A) 1 B) 2 C) 4 D) 8

65. $P(13) - Q(13)$ is equal to

A) 81 B) 78 C) 91 D) 65

Passage – 3 (Q. No. 66 to 67)

There are two die D_1 and D_2 both having six faces D_1 has 3 faces marked with 1, 2 faces marked with 2 and 1 face marked with 3. D_2 has 1 face marked with 1, 2 faces marked with 2 and 3 faces marked with 3. Both dice are thrown once. Let $P(x)$ be the probability of getting the sum equal to x. Then

66. $P(3) =$

A) $\frac{1}{12}$ B) $\frac{1}{9}$ C) $\frac{5}{36}$ D) $\frac{2}{9}$

67. Which is the correct order

A) $P(2)<P(6)<P(3)<P(4)<P(5)$ B) $P(6)=P(2)<P(3)=P(5)<P(4)$

C) $P(6)=P(2)<P(5)<P(3)<P(4)$ D) $P(6)=P(2)<P(3)<P(5)<P(4)$

Passage – 4 (Q.No. 68 to 70)

A chess match between two grand master X and Y is won by whoever first wins a total of two games. X's probability of winning, drawing or losing any particular game are *p, q, r* respectively. The games are independent and $p+q+r=1,$ then

68. The probability that X wins the match after (n) games $(n\geq 5)$ is

A) $np^2q^{n-3}(q+(n-1)r)$ B) $(n-1)p^2q^{n-2}$

C) $p^2(n-2)q^{n-3}(q+(n-1)r)$ D) $p^2q^{n-3}(n-1)(q+(n-2)r)$

69. The probability that Y wins the match after 6^{th} game

A) $5q^3r^2(q+4p)$ B) $6q^3r^2(q+5p)$ C) $4q^3r^2(q+5p)$ D) $5q^4r^2$

70. The probability that Y wins the match is

A) $\dfrac{r^2(3p+r)}{(1-q)^3}$ B) $\dfrac{r^2(2p+r)}{(1-q)^3}$ C) $\dfrac{r^2(p+3r)}{(1-q)^3}$ D) $\dfrac{r^2(p+2r)}{(1-q)^3}$

Passage – 5 (Q.No. 71 to 72)

Let $\Delta=\begin{vmatrix}(10^4+2) & (10^7+3) & (10^8+8)\\ (10^9+9) & (10^2+8) & (10^3-4)\\ (10^3-5) & (10^8+b) & (10^6+a)\end{vmatrix}$, where $a,b\in\{1,2,3,4,5,6,7,8,9\}$. Then

71. The probability that Δ is odd is equal to

A) $\dfrac{4}{9}$ B) $\dfrac{5}{9}$ C) $\dfrac{40}{81}$ D) $\dfrac{25}{81}$

72. Let $k\in N$ such that the least prime factor of *k* is 7 and $a=3,b=2.$ Then the least prime factor of $k+\Delta$ is

A) 2 B) 3 C) 5 D) 7

Passage – 6 (Q.No. 73 to 74)

Consider two 3×3 matrices A and B satisfying $A=adj(B)-B^T$ and $B=adj(A)-A^T$. Also given that A is a non singular matrix.

(Where P^T denotes transpose of matrix P).

73. $|A|+|B|$ is equal to

A) 4 B) 8 C) 16 D) 128

74. $AB+BA$ is equal to

A) 161 B) 21 C) 41 D) 81

Passage – 7 (Q. No. 75 to 77)

On the sides AB and BC of ΔABC, squares are drawn with centres D and E respectively such that points C and D lie on the same side of line AB and the point A and E lie on opposite sides of line BC. A, B, C are represented by complex number $1, \omega, \omega^2$ respectively $\left(\omega = \dfrac{-1+i\sqrt{3}}{2}\right)$

75. Angle between AC and DE is

A) $\frac{\pi}{6}$ B) $\frac{\pi}{4}$ C) $\frac{\pi}{3}$ D) $\frac{\pi}{2}$

76. The length of DE is

A) $\sqrt{\frac{3}{2}}$ B) $\frac{\sqrt{3}}{2}$ C) $\sqrt{3}$ D) $\sqrt{6}$

77. The length of AE is

A) $\frac{3+\sqrt{3}}{4}$ B) $\frac{3-\sqrt{3}}{2}$ C) $\frac{3-\sqrt{3}}{4}$ D) $\frac{3+\sqrt{3}}{2}$

Passage – 8 (Q. No. 78 to 80)

Let the point $P(z)$ lies on the curve 'C' whose equation is given by $\left|z+\frac{1}{z}\right|=2$ in the argand plane. If 'O' represents origin, then

78. The largest value of OP is equal to

A) $\sqrt{5}$ B) $\sqrt{2}$ C) $1+\sqrt{5}$ D) $1+\sqrt{2}$

79. If $\arg(z)=\frac{\pi}{6}$, then the sum of all possible value of $|z|$ is equal to

A) $\sqrt{5}$ B) $\sqrt{2}$ C) 2 D) $\sqrt{3}$

80. 'C' in argand plane represents

A) Two concentric circles B) Two orthogonal circles
C) Two circles non intersecting D) Two externally touching circles

Matrix Matching type

81. Let A be the matrix of order 3 such that $A=\left[a_{ij}\right]_{3\times 3}$, where $a_{ij}=\begin{cases}0 & for\ i=j\\ 1 & for\ i\neq j\end{cases}$, then

Column – I		Column – II	
A)	$\left\|A^{-1}+I\right\|=$	P)	0
B)	$\left\|A^{3}-A^{2}+2A\right\|=$	Q)	4
C)	$\left\|A^{4}-4A-7I\right\|=$	R)	8
D)	$\left\|A^{4}-8A^{-1}-A^{2}-6I\right\|=$	S)	64

82. Let $A(t)=\left[a_{ij}\right]$ is a matrix order 3×3 given by $a_{ij}=\begin{cases}2\cos t & if\ \ i=j\\ 1 & if\ \left|i-j\right|=1,\\ 0 & otherwise\end{cases}$ then

Column – I		Column – II	
A)	The number of t interval $\left[-2\pi,4\pi\right]$ such that $\left\|A(t)\right\|=4$ is equal to	P)	0
B)	$\left\|A\left(\frac{\pi}{17}\right)\right\|\left\|A\left(\frac{4\pi}{17}\right)\right\|$ is equal to	Q)	1
C)	The maximum value of $\left\|A(t)\right\|\left\|A(2t)\right\|,\ \forall t\in R$ is equal to	R)	4
D)	$\int_0^{\pi}\left\|A(t)\right\|\left\|A(4t)\right\|dt$ is equal to	S)	6
		T)	8

83. Let A, B are two square matrices such that

Column – I		Column – II	
A)	A is non singular and AB = 0, then	P)	A = 0
B)	A and B both are not null matrices satisfy AB=0, then	Q)	B = 0
C)	$A^{n}=0$ for some $n\geq 2, n\in N$ then	R)	$\left\|A\right\|=0$
D)	B is non singular and Ab = 0, then	S)	$\left\|B\right\|=0$
		T)	$\left\|A\right\|+\left\|B\right\|\neq 0$

84. Match the column

Column – I		Column – II	
A)	Let A be a square matrix which commutes with a square matrix B of same order (B is not null matrix). If $AA^T = I$ and $BA^T - A^T B = kI$, then k is equal to	P)	0
B)	If C is a 3×3 skew symmetric matrix and X is any column vector, then $X^T CX = kI$, where k is equal to	Q)	1
C)	Let A be an $n\times n$ square matrix given by $A=\begin{bmatrix} 0 & 1 & 1 & & \ldots\ldots & 1 \\ 1 & 0 & 1 & & \ldots\ldots & 1 \\ 1 & 1 & 0 & 1 & \ldots\ldots & 1 \\ \vdots & \vdots & \vdots & & & \vdots \\ & & & & & 1 \\ 1 & 1 & 1 & & \ldots\ldots & 0 \end{bmatrix}$, then in the inverse of A, each diagonal element is equal to $\frac{\lambda_1 - n}{n-\lambda_2}$, then $\lambda_1 =$	R)	2
D)	In part (C), $\lambda_2 =$	S)	-1
		T)	3

85. Let A, B, C, D be non singular matrices of order 3×3 such that $\left(AA^T\right)BC^2 = B^T$ and $C^2 = I$ and D is an orthogonal matrix such that A^T and D commute with each other and $D^T A^T D = A^{-1}$. Then

Column – I		Column – II	
A)	If $B = B^{-1}$, them $\left\|B^{2018}C\right\|$ be equal to	P)	-1
B)	If $\left(ACA^T\right)^{2019} = AC^k A^T$, then k can be equal to	Q)	0
C)	$\left\|ACA^T\right\|$ can be equal to	R)	1
D)	$\left\|BB^T\left(B^{-1}\right)^2\right\| + \left\|\left(AA^T\right)^{2018}\right\|$ is equal to	S)	2
		T)	3

86. Match the column

Column – I		Column – II	
A)	If the roots of equation $x^3 - 9x^2 + 26x - k = 0$ are positive and in A.P., then k is equal to	P)	7
B)	If the roots of equation $x^3 - 14x^2 + kx - 64 = 0$ are positive and in G.P., then k is equal to	Q)	11
C)	If the roots of equation $6x^3 - kx^2 + 6x - 1 = 0$ are positive and in H.P., then k is equal to	R)	24
D)	In the equation $x^3 - kx + 6 = 0,$ the sum of two roots is 3, then the value of k is equal to	S)	26
		T)	56

87. Match the column

Column – I		Column – II	
A)	If $A = \sum_{m=1}^{n}\sum_{r=1}^{m} r - \sum_{r=1}^{n} r,$ then $\frac{A}{B}$ is equal to	P)	1
B)	For positive numbers a, b, c minimum value of $\frac{a\left(b^2+c^2\right)+b\left(c^2+a^2\right)+c\left(a^2+b^2\right)}{abc}$ is equal to	Q)	2
C)	If $x + y + z = 1,\ x, y, z > 0$, then the minimum value of $\frac{2x^2}{y+z} + \frac{2y^2}{z+x} + \frac{2z^2}{x+y}$ is equal to	R)	3
D)	If $\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx} = 3$ where $x, y, z \in N,$ then $x + y + z$ is equal to	S)	4
		T)	6

88. Match the column

Column – I		Column – II	
A)	If $\log_2 x + 4\log_4 y = 4 - \log_8 z,$ then $x + y + z$ can be equal	P)	2
B)	Let $x^2 - 3x + p = 0$ has two positive roots 'a' and 'b', then $\frac{4}{a} + \frac{1}{b}$ can be equal to	Q)	3
C)	For triangle ABC, if $\operatorname{cosec} A, \operatorname{cosec} B, \operatorname{cosec} C$ are in H.P., then possible value(s) of $\frac{2b}{c}$ will be (where a, b, c denote lengths of sides of ΔABC as in usual notation)	R)	5

D)	If $3^{\lvert\sin x\rvert}2^{-\lvert\sec y\rvert} = a - 5\cos z,\ x, y, z \in R$ and $y \neq (2n+1)\dfrac{\pi}{2}, n \in I$, then possible value(s) of a can be	S)	6
		T)	7

89. Match the column

Column – I		Column – II	
A)	Let a,b,c are positive real numbers such that $a^3b^2c = 12$, then the minimum value of $49a + 3b + c$ is equal to	P)	1
B)	The minimum value of $\left\lvert 2x^3 - \dfrac{3}{x^2} \right\rvert$ for $x < 0$ equal to	Q)	5
C)	The maximum value of $\dfrac{x^5\left(8 - x^3\right)}{\sqrt[3]{25}}$ for $0 < x < 2$ is equal to	R)	7
D)	If $x^7y^5 = a$ and $7x + 5y \geq 12\quad \forall x, y > 0$, then the minimum value of 'a' is equal to	S)	15
		T)	42

90. Consider a sequence $\{b_n\}$ of integers such that b_1, b_2, b_3 are in G.P. b_2, b_3, b_4 are in A.P., b_3, b_4, b_5 are in G.P., b_4, b_5, b_6 are in A.P., b_5, b_6, b_7 are in G.P. and so on. Also given that $b_1 = 1$ and $b_5 + b_6 = 198$. Then

Column – I		Column – II	
A)	$\sqrt{b_7}$ is equal to	P)	5
B)	Sum of digits of b_8 is equal to	Q)	15
C)	$\sqrt{b_9}$ is equal to	R)	9
D)	Sum of digits of b_{10} is equal to	S)	17
		T)	13

Single answer Type questions

91. If $\log_4 5 = a$ and $\log_5 6 = b$, then $\log_3 2$ is equal to

A) $2ab + 1$ B) $\dfrac{1}{2ab - 1}$ C) $\dfrac{1}{2ab + 1}$ D) $\dfrac{1}{ab - 1}$

92. Let (x, y) be the solution of following equation $\left[5(x+1)\right]^{\ln 5} = (2y)^{\ln 2}$ and $(x+1)^{\ln 2} = 5^{\ln y}$, then x is equal to

A) $\dfrac{1}{5}$ B) $-\dfrac{1}{5}$ C) $\dfrac{4}{5}$ D) $-\dfrac{4}{5}$

93. The value of ab if $\log_8 a+\log_4 b^2=5$ and $\log_8 b+\log_4 a^2=7$ is equal to

A) 64 B) 256 C) 512 D) 1024

94. Let x, y, z and w be positive number such that $\log_x w=24,\ \log_y w=40$ and $\log_{xyz} w=12$, then $\log_z w=$

A) 120 B) 40 C) 80 D) 60

95. The value of $\log_2 x$ if $\log_2(\log_8 x)=\log_8(\log_2 x)$ is equal to

A) 27 B) $3\sqrt{3}$ C) 3 D) $\sqrt{3}$

96. The number $\dfrac{2}{\log_4(2000)^6}+\dfrac{3}{\log_5(2000)^6}$ is equal to

A) $\dfrac{1}{6}$ B) $\dfrac{1}{3}$ C) $\dfrac{1}{2}$ D) 1

97. If $\log_6 a+\log_6 b+\log_6 c=6,\ a,\ b$ and c are positive integers that form a increasing G.P. and $b-a$ is the square of an integer. Then $a+b+c=$

A) 97 B) 101 C) 109 D) 111

98. Suppose $x\in\left[0,\dfrac{\pi}{2}\right]$ and $\log_{(24\sin x)}(24\cos x)=\dfrac{3}{2}$. Then $\cot^2 x=$

A) 3 B) 4 C) 6 D) 8

99. If $\log_b n=2$ and $\log_n(2b)=2$, then the value of $\log_{b^2}(2b)=$

A) 0 B) 1 C) 2 D) 4

100. The number of real values of x satisfying the equation $\log_{10}(\log_{10} x)+\log_{10}\left[\log_{10}\left(x^3\right)-2\right]=0$ is

A) 0 B) 1 C) 2 D) 3

KEY

1) BCD	2) ABD	3) CD	4) AB	5) ABD	6) ABC	7) BCD	8) BC
9) BC	10) AB	11) ACD	12) ABC	13) BC	14) ACD	15) BCD	16) BCD
17) CD	18) ABC	19) BCD	20) BCD	21) BD	22) AD	23) BD	24) ABD
25) ABC	26) ABD	27) BCD	28) BCD	29) AD	30) BC	31) ABD	32) CD
33) AD	34) AB	35) AB	36) BD	37) AD	38) CD	39) ABD	40) 2
41) 9	42) 7	43) 5	44) 2	45) 3	46) 0	47) 1	48) 7
49) 3	50) 1	51) 54	52) 9	53) 5	54) 0	55) 101	56) 21
57) 47	58) 46	59) 216	60) D	61) B	62) A	63) D	64) B
65) C	66) D	67) B	68) D	69) A	70) A	71) B	72) A
73) C	74) D	75) B	76) A	77) D	78) D	79) A	80) B
81) A – P; B – T; C – Q; D – R				82) A – R; B – Q; C – T; D – P			
83) A – Q,S,T; B – R,S; C – R; D – P,R,T				84) A – P; B – P; C – R; D – Q			
85) A – P,R; B – R,T; C – P,R; D – S				86) A – R; B – T; C – Q; D – P			
87) A – Q; B – T; C – P; D – R				88) A – R,S,T; B – Q,R,S,T; C – P,Q; D – P,Q,R,S			
89) A – T; B – Q; C – S; D – P				90) A – T; B – P; C – S; D – Q			
91) B	92) D	93) C	94) D	95) B	96) A	97) D	98) D
99) C	100) B						

HINTS

1. $$y=\frac{\tan 3x}{\tan x}=\frac{3-\tan^2 x}{1-3\tan^2 x}=\frac{8}{3(1-3\tan^2 x)}+\frac{1}{3}$$

$$\Rightarrow y\in\left(-\infty,\frac{1}{3}\right)\cup(3,\infty)$$

$$a=\frac{1}{3},\ b=3,\ S_\infty=\frac{6}{1-\frac{1}{3}}=9$$

2. $$a_{n+1}+1=\left(\sqrt{1+a_n}+1\right)^2$$

$$\sqrt{1+a_{n+1}}=\sqrt{1+a_n}+1$$

$\left\{\sqrt{1+a_n}\right\}$ is an A.P. with first term $\sqrt{1+a_1}=1$ and common difference 1.

$$\sqrt{1+a_1}=1+(n-1)1=n\Rightarrow a_n=n^2-1$$

3. $$\frac{a_1+2}{2}=\sqrt{2a_1}\Rightarrow\left(a_1-2\right)^2=0\Rightarrow a_1=2$$

$$\left(a_n-2\right)^2=8S_n$$

$$(a_{n-1}-2)^2 = 8S_{n-1}$$
$$(a_n+2)^2-(a_{n-1}-2)^2 = 8a_n$$
$$(a_n-a_{n-1})(a_n+a_{n-1}+4) = 8a_n$$
$$a_n^2-a_{n-1}^2 = 4(a_n-a_{n-1})$$
$$a_n-a_{n-1} = 4$$

4. $$\frac{3x+\frac{4xz}{x+z}}{2x-\frac{2xz}{x+z}}+\frac{3z+\frac{4xz}{x+z}}{2z-\frac{2xz}{x+z}} = \frac{3x^2+7xz}{2x^2}+\frac{3z^2+7xz}{2z^2}$$
$$=3+\frac{7}{2}\left(\frac{z}{x}+\frac{x}{z}\right)\in(10,\infty)$$

5. $$\frac{5^{a_{n+1}}}{5^{a_n}} = \frac{3n+5}{3n+2}$$
$$\frac{5^{a_2}}{5^{a_1}}\times\frac{5^{a_3}}{5^{a_2}}\times\ldots\ldots\times\frac{5^{a_n}}{5^{a_{n-1}}} = \frac{8}{5}\times\frac{11}{5}\times\frac{14}{11}\times\ldots\ldots\times\frac{3n+2}{3n-1}$$
$$\frac{5^{a_n}}{5^{a_1}} = \frac{3n+2}{5}\Rightarrow 5^{a_n} = 3n+2$$
$$[a_n]=3 \quad \Rightarrow \log_5(3n+2)\in[3,4)$$
$$\Rightarrow n\in\{41,42,\ldots\ldots..,207\}$$
$$[a_n]=4 \;\Rightarrow \frac{623}{3}\le n<\frac{3123}{3}$$
$$\Rightarrow n\in\{208,209,\ldots\ldots..,1041\}$$

6. $(a)\quad \Rightarrow (1.3.5\ldots(2n-1))^{1/n} < \dfrac{1+3+5+\ldots(2n-1)}{n} = n$

$$\Rightarrow (1.3.5\ldots(2n-1)) < n^1$$

$(b)\ \dfrac{1+2+2^2+\ldots..+2^{n-1}}{n} > (2^{1+2+\ldots.(n-1)})^{1/n} = 2^{\frac{n-1}{2}}$

$$\Rightarrow 2^n > 1+n\,2^{\frac{n-1}{2}}$$

$(c)\ \dfrac{1}{n+1}+\dfrac{1}{n+1}+\ldots..+\dfrac{1}{n+1} > \dfrac{1}{2n}+\dfrac{1}{2n}+\ldots.+\dfrac{1}{2n} = \dfrac{1}{2}$

7. $$\frac{1}{a_{n+1}-1} = \frac{1}{a_n(a_n-1)} = \frac{1}{a_n-1}-\frac{1}{a_n}$$
$$\frac{1}{a_n-1}-\frac{1}{a_{n+1}-1} = \frac{1}{a_n}$$
$$\Rightarrow \sum_{r=1}^{2018}\frac{1}{a^r} = \frac{1}{a_1-1}-\frac{1}{a_{2019}-1} = 1-\frac{1}{a_{2019}-1} < 1$$
$$a_{2019}-1 = a_{2018}(a_{2018}-1) = a_{2018}a_{2017}(a_{2017}-1)$$
$$a_{2019}-1 = a_{2018}a_{2017}a_{2016}\ldots.a_2a_1(a_1-1)$$
$$= a_1a_2a_3\ldots\ldots a_{2018}$$

$$2018 < \frac{2018}{\frac{1}{a_1}+\frac{1}{a_2}+.....+\frac{1}{a_{2018}}} < (a_1a_2a_3......a_{2018})^{\frac{1}{2018}}$$

$$a_{2019}-1 > \frac{1}{(2018)^{2018}}$$

$$1-\frac{1}{a_{2019}-1} > 1-\frac{1}{(2018)^{2018}}$$

8. $$\frac{k}{(k-1)^{4/3}+k^{4/3}+(k+1)^{4/3}} < \frac{k}{(k-1)^{4/3}+((k-1)(k+1))^{2/3}+(k+1)^{4/3}}$$

$$=\frac{\left((k+1)^{2/3}-(k-1)^{2/3}\right)k}{(k+1)^2+(k-1)^2}$$

$$=\frac{1}{4}\left((k+1)^{2/3}-(k-1)^{2/3}\right)$$

$$S_n < \frac{1}{4}\sum_{k=1}^{n}\left((k+1)^{2/3}-(k-1)^{2/3}\right) = \frac{1}{4}\left((n+1)^{2/3}+n^{2/3}-1\right)$$

$$S_{999} < \frac{1}{4}\left((1000)^{2/3}+(999)^{2/3}-1\right) < \frac{1}{4}(100+100-1) < 50$$

$$S_{999} < \frac{1}{4}\left((1000)^{2/3}+(999)^{2/3}-1\right) < \frac{1}{4}(100+100-1) < 50$$

$$S_{999} < \frac{1}{4}\left((27)^{2/3}+(26)^{2/3}-1\right) < \frac{1}{4}(9+9-1) < \frac{17}{4}$$

9. $$S=\frac{1}{\sqrt{1}+\sqrt{3}}+\frac{1}{\sqrt{5}+\sqrt{7}}+....+\frac{1}{\sqrt{9997}+\sqrt{9999}}$$

$$>\frac{1}{\sqrt{3}+\sqrt{5}}+\frac{1}{\sqrt{7}+\sqrt{9}}+....+\frac{1}{\sqrt{9999}+\sqrt{10001}}$$

$$2S > \frac{1}{\sqrt{1}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{5}}+\frac{1}{\sqrt{5}+\sqrt{7}}+....+\frac{1}{\sqrt{9999}+\sqrt{10001}}$$

$$=\frac{1}{2}\left(\sqrt{3}-\sqrt{1}+\sqrt{5}-\sqrt{3}+\sqrt{7}-\sqrt{5}+....+\sqrt{10001}-\sqrt{9999}\right)$$

$$=\frac{1}{2}\left(\sqrt{10001}-\sqrt{1}\right) > \quad =\frac{1}{2}(100-1)$$

$$\Rightarrow S > \frac{99}{4} > 24$$

10. $a_9 = 5a_2 \Rightarrow a_1+8d = 5(a_1+d) \Rightarrow 4a_1 = 3d$

$a_{13} = 2a_6+5 \Rightarrow a_1+12d = 2(a_1+5d)+5 \Rightarrow a_1+5 = 2d$

$\Rightarrow a_1 = 3, d = 4$

11. $1\ digit\ plaindromes \rightarrow 9$

$2\ digit\ plaindromes \rightarrow 9\times 1$

$3\ digit\ plaindromes \rightarrow 9\times 10$

$4\ digit\ plaindromes \to 9\times 10$

$5\ digit\ plaindromes \to 9\times 10^2$

$6\ digit\ plaindromes \to 9\times 10^2$

$1\,\underline{0}\,\underline{0} ------- \to 10$

$1\,0\,1 ------- \to 10$

$\therefore number\ at\ 2(9+90+900+10)=2018$, $rank\ is\ 10$

$1 ---------- \to 10^3$

$2\,0 -------- \to 10^2$

$2\,1\,0 ------- \to 10$

$2\,1\,1 \to 10$

$2\,1\,2 \to 10$

$2\,1\,3 \to 10$

$2\,1\,4\,0\,4\,1\,2 \to 1$

$$\begin{aligned} rank\ of\ 2140412 &= 2(9+90+900)+1000+100+40+1 \\ &= 1998+1000+100+40+1 \\ &= 3139 \end{aligned}$$

12. number of ways = $= \dfrac{{}^{2n}C_1\ {}^{2n-3-(k-1)+1}C_{k-1}}{k}$

$$= \frac{2n}{k}\ {}^{2n-1-k}C_{k-1}$$

$$or\quad {}^{2n-k+1}C_k - {}^{2n-k-1}C_{k-2}$$

13. If x is odd

x=2k+1, $y=2k+1,\ z=2k+1,2k+2,.....98-4k \to (98-6k)\ ways$

$y=2k+2,\ z=2k+2,\ 2k+3,.....97-4k \to (96-6k)\ ways$

$y=49-k,\ z=49-k,\ 50-k \to 2\ ways$

$$\therefore number\ of\ triplets = \sum_{k=0}^{16}\left((98-6k)+(96-6k)+...+2\right)$$

$$= \sum_{k=0}^{16}\left((49-3k)(50-3k)\right) = 14722$$

If x is even

x=2k+1 $y=2k,\ z=2k,\ 2k+1,.....,100-4k \to (101-6k)\ ways$

$y=2k+1,\ z=2k+1,\ .....99-4k \to (99-6k)\ ways$

$\vdots$

$y=50-k,\ z=50-k \to 1\ ways$

$$\therefore number\ of\ triplets = \sum_{k=0}^{16}\left((101-6k)+(99-6k)+...+1\right)$$

$$= \sum_{k=0}^{16}\left((51-3k)^2\right) = 16065$$

14. $x+y=9\times 10\times 10\times 10\times 10\times 10 = 9\times 10^5$

$$x=y=\frac{1}{2}\times 9\times 10^5 = 450000$$

15. $a_n = a_{n-1}+n$

$a_1 = 2$

$\therefore a_6 = 2+2+3+4+5+6 = 22$

$a_{10} = 2+2+3+4+....+10 = 56$

16. $a_0 + a_2 + a_4 + ...+ a_{38} + a_{40} = \dfrac{4^{20}+2^{20}}{2} = 2^{19}(2^{20}+1)$

$a_{40} = a_{20}$

$N = 2^{19}(2^{20}-1)$

17. $Put\ x = i \quad \Rightarrow i^n = (a_0 - a_2 + a_4 - a_6 + ...) + i(a_1 - a_3 + a_5 - a_7 + ...)$

$$\Rightarrow a_1 - a_3 + a_5 - a_7 + ... = \begin{cases} 0 & if\ n\ is\ even \\ 0 & if\ n = 4k+1, k \in 1 \\ -1 & if\ n = 4k+3 \end{cases}$$

18. $(13^n - 13^n) + 7^n = 10k + 7^n$

$\Rightarrow last\ digit\ is\ 3\ if\ n = 4k+3, k \in 1$

19. $2^{4n} - 2^n(7n+1) = 2^n\left[(1+7)^n - 7n - 1\right]$

$= 2^n\left({}^nC_2 7^2 + {}^nC_3 7^3 + ...\right)$

20. $S_n = \sum_{k=0}^{n} \dfrac{{}^{n+k}C_n}{2^k}$

$$= coefficient\ of\ x^n\ in \left((1+x)^n + \frac{(1+x)^{n+1}}{2} + \frac{(1+x)^{n+2}}{2^2} + ... + \frac{(1+x)^{n+n}}{2^n}\right)$$

$$= coefficient\ of\ x^n\ in\ \frac{(1+x)^n\left(\left(\frac{1+x}{2}\right)^{n+1} - 1\right)}{\left(\frac{1+x}{2} - 1\right)}$$

$$= coefficient\ of\ x^n\ in\ \frac{1}{2^n}\frac{\left((1+x)^{2n+1} - 2^{n+1}(1+x)^n\right)}{x-1}$$

$$= coefficient\ of\ x^n\ in\ \frac{1}{2^n}\left(2^{n+1}(1+x)^n - (1+x)^{2n+1}\right)\left(1 + x + x^2 + ...\infty\right)$$

$$= \frac{1}{2^n}\left[2^{n+1}\left({}^nC_0 + {}^nC_1 + {}^nC_2 + {}^nC_3 + ... + {}^nC_n\right) - \left({}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + ... + {}^{2n+1}C_n\right)\right]$$

$$= \frac{1}{2^n}\left[2^{n+1} - \frac{1}{2}2^{n+1}\right] = 2^n$$

21. $coefficient\ of\ x^1 = {}^tC_t + {}^{t+1}C_t + {}^{t+2}C_t + ... + {}^nC_t = {}^{n+1}C_{t+1}$

$coefficient\ of\ x^{t+1} = {}^{t+1}C_{t+1} + {}^{t+2}C_{t+1} + {}^{t+3}C_{t+1} ..... + {}^nC_{t+1} = {}^{n+1}C_{t+2}$

${}^{n+1}C_{t+1} = {}^{n+1}C_{t+2} \Rightarrow n - t = t + 2$

$\Rightarrow t = \dfrac{n-2}{2}\ their\ sum = {}^{n+2}C_{t+2}$

22. (a) tr(A) may be equal to 0

(b) $A^2 = 1 \Rightarrow A^2 + A^4 + ... + A^{50} = 251$

$\Rightarrow \left|A^2 + A^4 + ... + A^{50}\right| = (25)^n$

(c) $adj(|A|A) = |A|\,adj(A)$

$|adj(|A|A)| = |A|^2\,|adj(A)| = |A|^2\,|A| = |A|^3$

23. $AB = A \Rightarrow ABA = A^2 \Rightarrow AB = A^2$

$BA = B \Rightarrow BAB = B \Rightarrow BA = B^2 \Rightarrow B = B^2$

$A^2 - AB + B^2 = B^2 = B$

24. $A = \begin{bmatrix} 0 & \frac{1}{2} & 1 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 1 & \frac{1}{2} & 0 \end{bmatrix}$

$|A - \lambda I| = 0 \Rightarrow \begin{bmatrix} -\lambda & \frac{1}{2} & 1 \\ \frac{1}{2} & -\lambda & \frac{1}{2} \\ 1 & \frac{1}{2} & -\lambda \end{bmatrix} = 0$

$\lambda^3 - \frac{3}{2}\lambda - \frac{1}{2} = 0$

$2A^3 - 3A - I = O$

25. $BA = B \Rightarrow BAB = B^2 \Rightarrow BA = B^2 \Rightarrow B = B^2$

$AB = A \Rightarrow ABA = A^2 \Rightarrow AB = A^2 \Rightarrow A = A^2$

26. $A = IC \Rightarrow C = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 4 & 2 & 0 \end{bmatrix}$

$(I + C)^{100} = I + {}^{100}C_1 C + {}^{100}C_2 C^2 + {}^{100}C_3 C^3 + \ldots\ldots + {}^{100}C_{100} C$

$C^2 = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 4 & 2 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 4 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$C^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 4 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow A^{100} = I + 100C + 4950C^2$

$B = A^{100} - I = \begin{bmatrix} 0 & 0 & 0 \\ 200 & 0 & 0 \\ 20200 & 200 & 0 \end{bmatrix}$

27. $iz^3 + z^3 - z + i = (z - i)(iz^2 - 1) = 0$

$\Rightarrow \qquad z = i, z^2 = -i$

$\Rightarrow \qquad z = i, \frac{1}{\sqrt{2}}(1-i), \frac{1}{\sqrt{2}}(-1+i)$

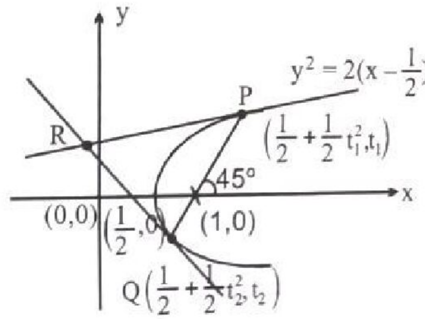
28. $x^2 = (x-1)^2 + y^2$

$\frac{2}{t_1+t_2} = 1$

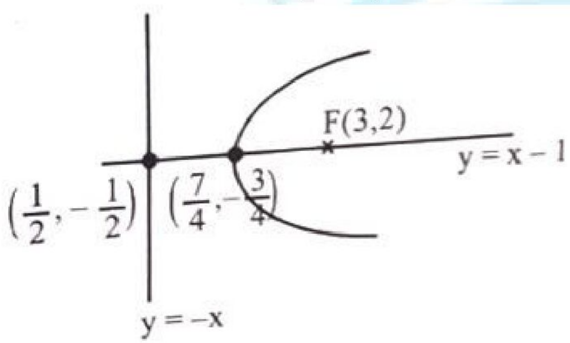
$\Rightarrow t_1 + t_2 = 2$

$R \equiv \left(\frac{1}{2}t_1t_2 + \frac{1}{2}, \frac{1}{2}(t_1+t_2)\right) = \left(\frac{1}{2} + \frac{1}{2}t_1t_2, 1\right)$

$\Rightarrow$ R lies on Im(z) = 1



29. $\sqrt{(x-3)^2 + (y-2)^2} = \left|\frac{x+y}{\sqrt{2}}\right|$



30. $z^n - 1 = (z-1)(z-z_1)(z-z_2)......(z-z_{n-1}) = 0$

$1 + z + z^2 + ... + z^{n-1} = 0 \quad \begin{cases} z_1 \\ z_2 \\ \vdots \\ z_{n-1} \end{cases}$

Replace $\qquad \frac{1}{3-z} = y \Rightarrow z = \frac{3y-1}{y}$

$\Rightarrow y^{n-1} + y^{n-2}(3y-1) + y^{n-3}(3y-1)^2 + ... + (3y-1)^{n-1} = 0$

$\Rightarrow y^{n-1}\left(1+3+3^2+....+3^{n-1}\right) - y^{n-2}\left(1+2.3+3.3^2+4.3^3+.....+(n-1)3^{n-2}\right)+....+(-1)^{n-1} = 0$

31. **<u>Case-I:</u>** Triangle is right angled at Z.

$\Rightarrow \frac{z^2-z}{z^3-z}$ is purely imaginary

$\Rightarrow \frac{1}{z+1} = ki \Rightarrow z+1-\frac{1}{k}i$

Z lies on line $x = -1, z \neq -1$

<u>Case-II:</u> Triangle is right angled at Z.

$\Rightarrow \frac{z-z^2}{z^3-z^2}$ is purely imaginary

$\Rightarrow \frac{1}{2} = ki \Rightarrow x = 0$

Case-III: Triangle is right angled at Z.

$\Rightarrow \frac{z - z^3}{z^2 - z^3}$ is purely imaginary

$\frac{1+z}{z}$ is purely imaginary

$\Rightarrow 1 + \frac{x}{x^2 + y^2} = 0$

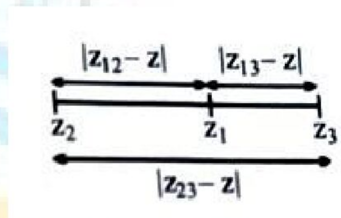
$\Rightarrow x(x+1) + y^2 = 0$

32. $(p - ip') = 4(q - ip')$

$\Rightarrow p^2 - (p')^2 = 4q \; and \; pp' = 2p'$

33. If A lies between B and C

$z_1 = \frac{|z_1 - z_2| z_1 + | z_1 - z_3 | z_2}{|z_2 - z_3|}$

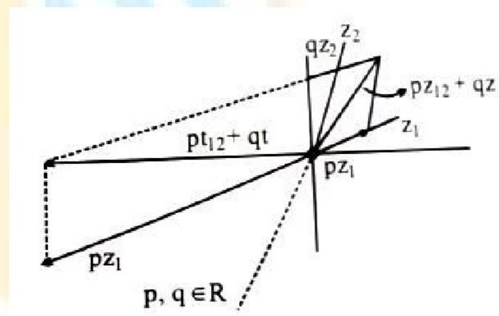


If A,B,C are non collinear excentre of ΔABC

$I_1 = \frac{-|z_2 - z_3| z_1 + |z_1 - z_2| z_3 + |z_1 - z_3| z_2}{-|z_2 - z_3| + |z_1 - z_2| + |z_1 - z_3|} = 0$

34. (a) Using ||gm law any complex no. z_3 in complex plane can be expressed as

$z_3 = pz_1 + pz_2$



(b)If z_1, z_2, z_3 are collinear, then

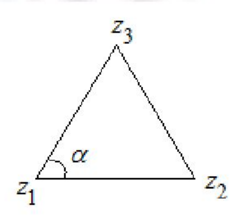
$z_3 = \frac{pz_1 + pz_2}{p+q}$

$\Rightarrow pz_1 + pz_2 - (p+q)z_3 = 0 \; and \; p + q + (-p - q) = 0$

(c) $\frac{z_2 - z_1}{|z_2 - z_1|} e^{i\alpha} = \frac{z_3 - z_1}{|z_3 - z_1|}$

$z_1 \left(|z_3 - z_1| e^{i\alpha} - |z_2 - z_1| \right) + z_2 \left(-|z_3 - z_1| e^{i\alpha} + z_3 |z_2 - z_1| \right) = 0$

And $|z_3 - z_1| e^{i\alpha} - |z_2 - z_1| - |z_3 - z_1| e^{i\alpha} + z_3 |z_2 - z_1| = 0$



35. Let $t = 3^{-|x-2|}$

$\Rightarrow t^2 - 4t = a, \; t(0,1]$

$\Rightarrow a \in [-3, 0)$

$t = 2 - \sqrt{4+a} \qquad \because t \le 1$

$$3^{-|x-2|} = 2-\sqrt{4+a}$$

$$\Rightarrow x = 2 \pm \log_3\left(2-\sqrt{4+a}\right)$$

36. $(x+y)\ln x = n\ln y$

$(x+y)\ln y = 2n\ln x + n\ln y$

Divide (2) by (1)

$$\frac{\ln y}{\ln x} = 2\frac{\ln x}{\ln y}+1 \Rightarrow \frac{\ln y}{\ln x} = 2,-1$$

$\frac{\ln y}{\ln x} = -1$ gives $(x,y)=(1,1)$

Taking $\ln y = 2\ln x \Rightarrow y = x^2$

37. $(2x+y)^2 = x^2+xy+7y^2$

$\Rightarrow x^2+xy-2y^2=0$

$\Rightarrow (x-y)(x+2y)=0$

Case-I: If $y=x,$ then

$$(3x+y)^2 = 3x^2+4xy+ky^2$$

$\Rightarrow 16y^2 = 7y^2+ky^2$

$\Rightarrow k=9$

Case-II : If $x=-2y$

$\Rightarrow (-5y)^2 = 12y^2-8y^2+ky^2$

$\Rightarrow k=21$

38. $a^{128}+128 = \log_a \log_a 2^{24}$

$$a^{\left(a^{a^{128}}.a^{128}\right)} = 2^{24}$$

$$a^{\left(a^{128}\right)^{a^{a^{128}}}} = 8^8$$

$\Rightarrow a^{a^{128}} = 8$

$\Rightarrow \left(a^{a^{128}}\right)^{128} = 8^{128}$

$\Rightarrow \left(a^{128}\right)^{a^{128}} = (64)^{64}$

$\Rightarrow a^{128} = 64$

$\Rightarrow a = 2^{\frac{3}{64}}$

$\Rightarrow 2^{\frac{3}{64}\times 256} = \log_{2^{\frac{3}{64}}} b$

$$\Rightarrow b = \left(2^{\frac{3}{64}}\right)^{2^{12}} = 2^{3\times 2^6} = 2^{192}$$

39. If $k<0$

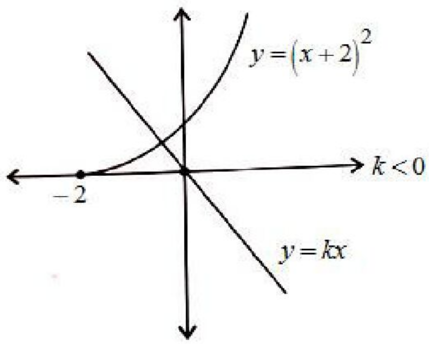
Equation has only one root

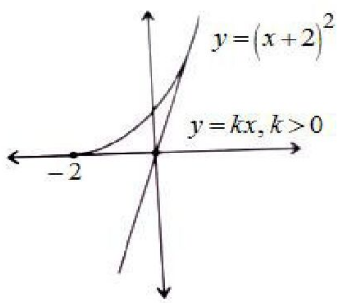
If $k>0$, $kx = x^2+4x+4$

$x^2+(4-k)x+4=0$

$D=0 \Rightarrow$ $(k-4)^2-16=0$

$\Rightarrow$ $k=8$





40. Let $f(x)=(ax+1)(x-a)>0, |x|\le 2$

Clearly a<0, otherwise $f(x)\to\infty$ as $x\to\infty$

Which is not required

$\therefore x\in\left(a,-\frac{1}{a}\right)$

$\therefore |a|\le 2$ and $\frac{1}{|a|}\le 2$

$\Rightarrow \frac{1}{2}\le |a|\le 2$

$\Rightarrow a\in\left[-2,-\frac{1}{2}\right]$

41. $8(x-1)p(x)=(x-8)p(2x)$

$\Rightarrow p(x)$ contain $x-8$ as factor

$\Rightarrow p(2x)$ contain $2(x-4)$ as factor

$\Rightarrow p(x)$ contain $(x-4)$ as factor

$\Rightarrow p(2x)$ contain $2(x-2)$ as factor

$p(x)$ contain $(x-2)$ as factor

$p(2x)$ contain $2(x-1)$ as factor

$$p(x)=(x-2)(x-4)(x-8)$$

$\Rightarrow p(10)=96$

42. $p(x)=(x-\alpha)g(x)$

Put $x=2,$ then $x=10,$ we get $g(x)$ integral will integral coefficients

$\Rightarrow 13=(2-\alpha)g(2)$

$\Rightarrow 2-\alpha=13,1,-13,-1$

and $5=(10-\alpha)g(5)$

$\Rightarrow 10-\alpha=5,1,-5,-1$

$\Rightarrow \alpha=15$

43. $1-2\sin^2 x+a\sin x=2a-7$

$\Rightarrow 2\sin^2 x-a\sin x+2a-8=0$

$\Rightarrow (\sin x-2)\left[2(\sin x+2)-a\right]=0$

$\Rightarrow$ $\sin x=2$ rejected and $\sin x=\dfrac{a-4}{2}$

$\therefore$ $-1\le\dfrac{a-4}{2}\le 1$

$\Rightarrow$ $a\in[2,6]$

44. $4^{\left|x^2-8x+12\right|=7^{2y}}$ $\Rightarrow y\ge 0$

If $0\le y\le 3$

$\Rightarrow 3-y-3y-2y^2-2-4y\ge 1$

$\Rightarrow$ $2y^2+8y\le 0$

$\Rightarrow$ $-4\le y\le 0$

$\Rightarrow$ $y=0,\ x=6,2$

If $y>3$

$\Rightarrow y-3-3y-2y^2-2-4y\ge 1$

$\Rightarrow$ $2y^2+6y+6\le 0$

$\Rightarrow$ $y^2+3y+3\le 0$

$\Rightarrow$ $y\in\phi$

Hence, $(x,y)=(6,0),(2,0)$

45. $D=(n+1)^2-4n(n+2)=-3n^2-6n+1=4-3(n+1)^2$

$D\ge 0\ \Rightarrow -\dfrac{2}{\sqrt{3}}\le n+1\le\dfrac{2}{\sqrt{3}}\Rightarrow n=-2,-,1,0$

For all values of n, D becomes perfect square

Hence, $n=-2,-1,0$

46.

$$ax^2+bx+c=0 \begin{array}{l} \nearrow \alpha-3d \\ \searrow \alpha+d \end{array}$$

$\Rightarrow$ $$2(\alpha-d)=-\frac{b}{a}$$

$$\alpha^2-3d^2-2\alpha d=\frac{c}{a}$$

$$cx^2+bx+a=0 \begin{array}{l} \nearrow \alpha-d \\ \searrow \alpha+3d \end{array}$$

$\Rightarrow$ $$2(\alpha+d)=-\frac{b}{c}$$

$$\alpha^2-3d^2+2\alpha d=\frac{a}{c}$$

$$4\alpha=-b\left(\frac{1}{a}+\frac{1}{c}\right), 4d=b\left(\frac{1}{a}-\frac{1}{c}\right)$$

$$4\alpha d=\frac{a}{c}-\frac{c}{a}=\frac{a^2-c^2}{ac}$$

Also $$16\alpha d=\frac{b^2\left(a^2-c^2\right)}{a^2c^2}$$

$\therefore$ $$\frac{a^2-c^2}{ac}=\frac{b^2\left(a^2-c^2\right)}{4a^2c^2}$$

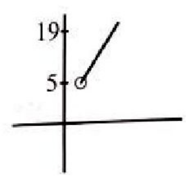
$\Rightarrow$ $a-c=0$ ro $a+c=0$ or $b^2=4ac$

$b^2=4ac$ and $a-c=0$ are rejected as p_1, q_1, p_2, q_2 are distinct.

Hence, $a+c=0$

47. $$\left(\frac{x^2+x+2}{x^2+x+1}\right)^2-(a-3)\frac{x^2+x+2}{x^2+x+1}+(a-4)=0$$

Put $\frac{x^2+x+2}{x^2+x+1}=t=1+\frac{1}{x+^2+x+1}, t\in\left(1,\frac{7}{3}\right]$



48. The number of points of intersection of the $A_iA_{i+9}=2\times 8=16$

Since, the lengths of these 200 diagonals are equal, they all are tangent to a same circle. Since from any point outside the circle exactly two tangents lines to the circle can be introduced, so any three of the 200 diagonals can't intersect at one common point.

$\Rightarrow$ Total number of interior point of intersection of 200 diagonals $=\frac{200\times 16}{2}=1600$

49. $N = {}^6C_3 + 4\,{}^6C_4 + 5\,{}^6C_5 + 6\,{}^6C_6 = 111$

$^6C_3$	$4\,^6C_4$	$5\,^6C_5$	$6\,^6C_6$
Using 3 points on the circle	Using 3 points on the circle and 1 interior point	Using 1 point on the circle and 2 interior point	Using all 3 interior Points

50. $1.......... = 5\times 5! = 600$

$2.......... = 5\times 5! = 600$

$3.......... = 5\times 5! = 600$

$41.......... = 4\times 4! = 96$

$42.......... = 4\times 4! = 96$

$431.......... = 3\times 3! = 18$

$4321.......... = 2\times 2! = 4$

$43251.......... = 2!$

$43256.......... = 2!$

2018^{th} number *is* 4325671

51. $(\log_a b)^2 - 5\log_a b + 6 = 0$

$\Rightarrow \log_a b = 2 \; or \; 3$

$\Rightarrow b = a^2 \; or \; a^3$

If $b = a^2 \Rightarrow a \le 44$

If $b = a^3 \Rightarrow a \le 12$

$\Rightarrow$ number of ordered pairs (a,b) = 43 + 11 =54.

52. Various forms of the numbers can be

$(i)\, abc15$ $(ii)\, ab150$ $(iii)\, ab155$ $(iv)\, a15b0$

$(v)\, a15b5$ $(vi)\, 15ab0$ $(vii)\, 15ab5$

Number of numbers of form $abc15 = 30\{102, 105, ..., 999\}$

Number of numbers of form $ab150 = 30\{13, 15, ..., 99\}$

Number of numbers of form $ab150 = 30\{10, 13, ..., 97\}$

Number of numbers of form $a15b0 = 30\{12, 15, ..., 99\}$

Number of numbers of form $a15b5 = 30\{10, 13, ..., 97\}$

Number of numbers of form $a15b0 = 34\{00, 03, ..., 99\}$

Number of numbers of form $a15b5 = 33\{01, 04, ..., 97\}$

53. $6\,A\; like \rightarrow 5$

$4\,A\; like + 2\, others\; alike \rightarrow 5\times 4\times \dfrac{6!}{4!2!} = 300$

$3\,A\; like + 3\, others\; alike \rightarrow {}^5C_2 \times \dfrac{6!}{4!2!} = 200$

$2\,A\,like + 2\,others\,alike + 2\,others\,alike \to {}^5C_3 \times \frac{6!}{2!2!2!} = 900$

$N = 900 + 200 + 300 + 5 = 1405$

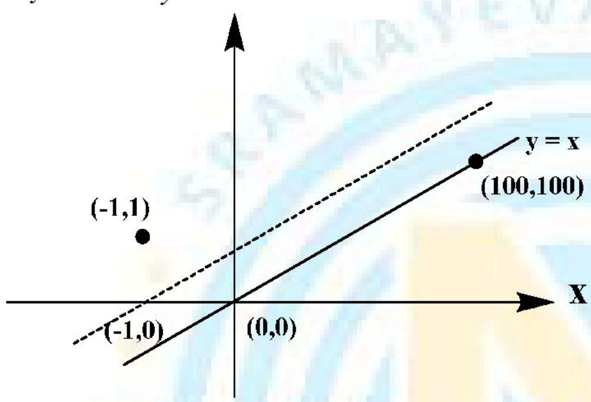
54. $012304 \quad \to \frac{6!}{2!} - 5!$

$012313 \quad \to \frac{6!}{2!2!} - \frac{5!}{2!2!}$

$012322 \quad \to \frac{6!}{3!} - \frac{5!}{3!}$

$N = 240 + 150 + 100 = 490$

55. Let till any point of time, there are $'x'$ Re notes and $'y'$ 200 Re notes. Then for not having any problem at any time $x \geq y$.



Shift origin to $(-1,0)$ and reflect $(0,0)$ about $y = x + 1$.

Favourable ways $=$ Total-going from $(\text{-}1,0)\,to\,(100,100)$

Favourable ways $= {}^{200}C_{100} - {}^{101+99}C_{101} = \frac{{}^{200}C_{100}}{101}$

$$P = \frac{\left(\frac{{}^{200}C_{100}}{101}\right)}{{}^{200}C_{100}} = \frac{1}{101}$$

56. $P = \frac{{}^6C_2\,{}^5C_3}{{}^{10}C_5} = \frac{5}{21}$

57. $P = \frac{{}^{30}C_{15}}{{}^{31}C_{15}} = \frac{16}{31}$

Alternate : $P = \frac{30}{31} \times \frac{14}{15} \times \frac{6}{7} \times \frac{2}{3} = \frac{16}{31}$

58. $a_1^{12} r^{66} = 8^{2006}$

$\Rightarrow a_1^2 r^{11} = 2^{1003}$

Let $a_1 = 2^x., r = 2$

$\Rightarrow 2x + 11y = 1003$

$$\Rightarrow y = \frac{1003 - 2x}{11}$$

$\Rightarrow x = 11k + 1, x \leq 496$

$\Rightarrow$ number of $x = 46$

59. $\left(\sqrt{x}\log_b x\right)^3 = b^{56}$

$(\log_b x)^{54} = x$

$$\Rightarrow \left(\sqrt{x}\, x^{\frac{1}{54}}\right)^3 = b^{56}$$

$\Rightarrow x^{\frac{14}{9}} = b^{56}$ $\qquad \Rightarrow x = b^{36}$

From (2), we get

$$(36)^{54} = x = b^{36}$$

$$\Rightarrow b = (36)^{3/2} = 216$$

Q.No. 60 to 62

$$2S_n = S_n - S_{n-1} + \frac{1}{S_n - S_{n-1}}$$

$$\Rightarrow 2S_n\left(S_n - S_{n-1}\right) = \left(S_n - S_{n-1}\right)^2 + 1 = S_n^2 + S_{n-1}^2 - 2S_nS_{n-1}$$

$$\Rightarrow S_n^2 = S_{n-1}^2 + 1 = \left(S_{n-2}^2 + 2\right) = \ldots\ldots\ldots$$

$$S_n^2 = S_1^2 + n - 1 = n \Rightarrow S_n = \sqrt{n}$$

Q.No. 63 to 65

$$\sum_{r=1}^{n}\left((r+1)^2 - r^2\right)\phi(r) = \sum_{r=1}^{n}(r+1)^2\left(\phi(r) - \phi(r+1)\right) + \sum_{r=1}^{n}\left((r+1)^2\phi(r+1) - r^2\phi(r)\right)$$

$$= -\sum_{r=1}^{n}(r+1) + (n+1)^2\phi(n+1) - 1^2\phi(1)$$

$$= -\left(1 + 2 + 3 + \ldots. + (n+1)\right) + (n+1)^2\phi(n+1)$$

$$= -\frac{(n+1)(n+2)}{2} + (n+1)^2\phi(n+1)$$

$$\sum_{r=0}^{9}P(r) = 1^2 + 2^2 + \ldots.10^2 = \frac{10 \times 11 \times 21}{6} = 385$$

$$\sum_{r=0}^{\infty}\frac{1}{Q(r)} = \sum_{r=0}^{\infty}\frac{2}{(r+1)(r+2)} = 2\sum_{r=0}^{\infty}\left(\frac{1}{r+1} - \frac{1}{r+2}\right) = 2$$

$$P(n) - Q(n) = (n+1)^2 - \frac{(n+1)(n+2)}{2} = \frac{n(n+1)}{2}$$

Q.No. 66 & 67

$$P(2)=\frac{3}{6}\times\frac{1}{6}=\frac{3}{36}$$
$$P(3)=\frac{3}{6}\times\frac{2}{6}+\frac{2}{6}\times\frac{1}{6}=\frac{8}{36}=P(1,2\ or\ 2,1)$$
$$P(4)=\frac{3}{6}\times\frac{3}{6}+\frac{1}{6}\times\frac{1}{6}+\frac{2}{6}\times\frac{2}{6}=\frac{14}{36}=P(1,3\ or\ 3,1\ or\ 2,2)$$
$$P(5)=\frac{2}{6}\times\frac{3}{6}+\frac{1}{6}\times\frac{2}{6}=\frac{8}{36}=P(2,3\ or\ 3,2)$$
$$P(6)=\frac{1}{6}\times\frac{3}{6}=\frac{3}{36}$$

Q.No. 68 to 70

Let $W\rightarrow Win, D\rightarrow Draw,\ L\rightarrow Lose\ \ for\ X$

$$P(X\text{ wins in n games})=P\left((2W,(n-2)D\ )or\ (2W,,1L,(n-3)D)\right)$$
$$={}^{n-1}C_1p^2q^{n-2}+(n-2)(n-1)p^2\,r\,q^{n-3}$$
$$=(n-1)p^2r^{n-2}+(n-1)(n-2)p^2q^{n-3}r$$
$$=(n-1)p^2q^{n-3}\left(q+(n-2)r\right)$$

$$P(Y\ wins)=P\left(2L,(4)D\right)\ or\ \left(2L,1W,3D\right)$$
$$={}^5C_1r^2q^4+5\times4\left(r^2pq^3\right)$$
$$=5q^3r^2(q+4p)$$
$$P(Y\ wins)=\frac{r^2}{(1-q)^2}+\frac{{}^2C_1pr^2}{(1-q)^3}$$
$$=\frac{r^2\left((p+r)+2p\right)}{(1-q)^3}=\frac{r^2(3p+r)}{(1-q)^3}$$

Q.No. 71 & 72

71. For Δ to be odd 'a' must be odd

$$\Rightarrow P=\frac{5}{9}$$

72. If a = 3, b = 5, then Δ is odd

$\Rightarrow k+\Delta$ is even

Q.No. 73 & 74

$$A+B^T=adjB$$
$$\Rightarrow A^T+B=(adj\,B)^t$$
and $B+A^T=adj\,A$
$$\Rightarrow (adj\,B)^T=adj\,A$$
$$\Rightarrow |B|^2=|A|^2 \qquad \Rightarrow |B|=\pm|A|$$

$$\left(|B|B^{-1}\right)^T = |A|A^{-1}$$

If $|B| = |A|$

$$\Rightarrow \left(B^{-1}\right)^T = A^{-1} = \left(B^T\right)^{-1}$$

$$\Rightarrow A = B^T$$

$$\therefore adj\, B = 2B^T \qquad \Rightarrow |B|^2 = 2^3|B|$$

$$\Rightarrow |B| = 8, |A| = 8$$

If $|B| = -|A|$

$$\Rightarrow \left(-B^{-1}\right)^T = A^{-1} = \left(-B^T\right)^{-1}$$

$$\Rightarrow A = -B^T$$

$\therefore adj\, B = O$ which is impossible

Now, $A = B^T$

$$adj\, B = 2A$$

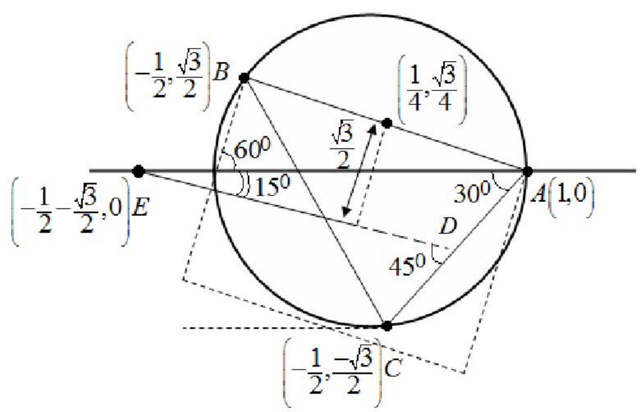
$$(adj\, B)B = 2AB = |B|I$$

$$\Rightarrow AB = 4I$$

Also $B\, adj\, B = 2BA = |B|I$

$$\Rightarrow BA = 4I$$

Q.No. 75 to 77



$$D \equiv \left(\frac{1}{4} - \frac{\sqrt{3}}{2}\frac{1}{2}, \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{2}\frac{\sqrt{3}}{2}\right)$$

$$= \left(\frac{1-\sqrt{3}}{4}, \frac{\sqrt{3}-3}{4}\right)$$

Slope of DE $= \dfrac{\frac{\sqrt{3}-3}{4}}{\frac{3}{4}+\frac{\sqrt{3}}{4}} = \sqrt{3}-2$

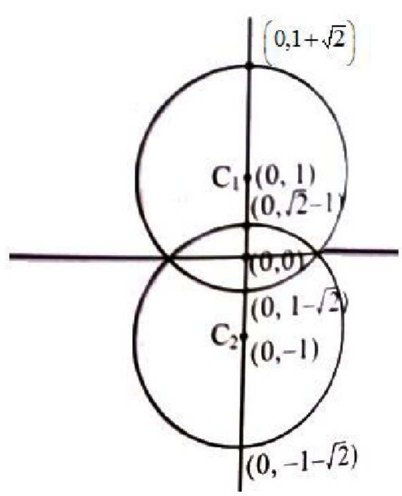
$\Rightarrow$ Angle between AC and DE $= \dfrac{\pi}{4}$

$$DE^2 = \left(\frac{\sqrt{3}-3}{4}\right)^2 + \left(\frac{\sqrt{3}-3}{4}\right)^2 = \frac{2}{16}(3+9) = \frac{3}{2}$$

$$AE = 1 + \frac{1}{2} + \frac{\sqrt{3}}{2} = \frac{3+\sqrt{3}}{2}$$

Q.No. 78 to 80

$$\left|z^2+1\right|^2 = 4|z|^2$$



$$\Rightarrow \quad (z^2+1)(\bar{z}^2+1) = 4|z|^2$$

$$\Rightarrow \left(|z|^4 - 2|z|^2 + 1\right) + \left(z^2 + \bar{z}^2 - 2|z|^2\right) = 0$$

$$\Rightarrow \left(|z|^2 - 1\right)^2 - \left(i(z-\bar{z})\right)^2 = 0$$

$$\Rightarrow \left(z\bar{z} - i(z-\bar{z}) - 1\right)\left(z\bar{z} + i(z-\bar{z}) - 1\right) = 0$$

$$|z|_{\max} = 1+\sqrt{2}$$

If $z = re^{i\frac{\pi}{6}}$, then

From $z\bar{z} - (z-\bar{z}) - 1 = 0$

$$r^2 + r - 1 = 0 \Rightarrow r = \frac{-1+\sqrt{5}}{2}$$

and from $z\bar{z} + i(z-\bar{z}) - 1 = 0$

$$r^2 - r - 1 = 0 \Rightarrow r = \frac{1+\sqrt{5}}{2}$$

Also $\quad C_1C_2^2 = 4 = r_1^2 + r_2^2 = \left(\sqrt{2}\right)^2 + \left(\sqrt{2}\right)^2$

$\Rightarrow$ Circles are orthogonal

81. $$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \Rightarrow |A| = 2, |A-I| = \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$A+I = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow |A+I| = 0$$

$$\Rightarrow (A+I)^2 = \begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{bmatrix} = 3(A+I)$$

$$\Rightarrow A^2 = A + 2I$$

$$A^3 = A^2 + 2A = 3A + 2I$$

$$A^4 = 3A^2 + 2A = 5A + 6I$$

$$A = I + 2A^{-1}$$

$$2A^{-1} = A - I$$

$$A^{-1} + I = \frac{1}{2}(A-I) + I = \frac{1}{2}(A+I)$$

$$A^3 - A^2 + 2A = 4A$$

$$\Rightarrow \left|A^3 - A^2 + 2A\right| = 4^3|A| = 128$$

$$A^4-4A-7I=A-I$$
$$\left|A^4-4A-7I\right|=\left|A-I\right|=4$$
$$A^4-8A^{-1}-A^2=8I$$
$$\Rightarrow A^4-8A^{-1}-A^2-6I=2I$$

82. $$A(t)=\begin{bmatrix}2\cos t & 1 & 0\\ 1 & 2\cos t & 1\\ 0 & 1 & 2\cos t\end{bmatrix}$$

$$\left|A(t)\right|=2\cos t\left(4\cos^2 t-1\right)-2\cos t=8\cos^3 t-4\cos t$$
$$\left|A(t)\right|=4\cos t\cos 2t$$

(A) $\left|A(t)\right|=4\Rightarrow t=2n\pi,\ n\in I$

$\Rightarrow t=-2\pi,0,2\pi,4\pi$

(B) $$\left|A\left(\frac{\pi}{17}\right)\right|\left|A\left(\frac{4\pi}{17}\right)\right|=16\cos\frac{\pi}{17}\cos\frac{2\pi}{17}\cos\frac{4\pi}{17}\cos\frac{8\pi}{17}$$
$$=\frac{\sin\frac{16\pi}{17}}{\sin\frac{\pi}{17}}=1$$

(C) $\left|A(t)\right|+\left|A(2t)\right|=4\cos t\cos 2t+4\cos 2t\cos 4t\le 8$

(D) $$\int_0^{\pi}16\cos t\cos 2t\cos 4t\cos 8t\ dt=\int_0^{\pi}\frac{\sin 16t\ dt}{\sin t}$$
$$\int_0^{\pi/2}\left(\frac{\sin 16t}{\sin t}+\frac{\sin\left(16\pi-16t\right)}{\sin\left(\pi-t\right)}\right)dt$$
$=0$

83. (A) $AB=O\Rightarrow A^{-1}AB=O\Rightarrow B=O$ and

(B) If $\left|A\right|\ne 0\Rightarrow B=O\ and\ if\ \left|B\right|\ne 0\Rightarrow A=O$

$\Rightarrow\left|A\right|=0\ and\ \left|B\right|=0$

(C) $A^n=O\Rightarrow\left|A\right|=0$

(D) $AB=O\Rightarrow ABB^{-1}=O\Rightarrow A=O$

84. (A) $AB=BA\Rightarrow B^TA^T=A^TB^T\ \Rightarrow\ AB^TA^T=B^T$

$\Rightarrow AB^T=B^TA\ \Rightarrow\ BA^T=A^TB$

(B) $\left(X^TCX\right)^T=X^TC^TX=-X^TCX$

$\Rightarrow\left|X^TCX\right|=-\left|X^TCX\right|\Rightarrow\left|X^TCX\right|=0$

$X^TCX=[0]$

(C) $A+I=\begin{bmatrix}1 & 1 & \cdots\cdots & 1\\ 1 & 1 & \cdots\cdots & 1\\ 1 & 1 & \cdots\cdots & 1\end{bmatrix}\Rightarrow(A+I)^2=n(A+I)$

$\Rightarrow A^2+(2-n)A+(1-n)I=0$

$\Rightarrow A^2+(2-n)I=(n-1)A^{-1}$

$\Rightarrow A^{-1}=\dfrac{A}{(n-1)}+\dfrac{2-n}{n-1}I$

$\lambda_1=2,\lambda_2=1$

85. $A^TD=DA^T$

$DD^T=D^TD=I$

$\Rightarrow DD^TA^TD=DA^T\Rightarrow DA^{-1}=DA^T\Rightarrow A^{-1}=A^T$

$\Rightarrow BC^2=B^T\Rightarrow B=B^T$

(A) $B=B^{-1}\Rightarrow B^2=I$

$\Rightarrow\left|B^{2018}C\right|=|C|=\pm1$

(B) $\left(ACA^T\right)^{2019}=\left(ACA^T\right)\left(ACA^T\right).....\left(ACA^T\right)=AC^{2019}A^T=AC^kA^T$

$\Rightarrow$ k is odd

(C) $\left|ACA^T\right|=|A|^2|C|=|C|=\pm1$

(D) $\left|BB^T\left(B^{-1}\right)^2\right|+\left|\left(AA^T\right)^{2018}\right|=1+1=2$

86. (A) $\alpha+\gamma=2\beta\Rightarrow\alpha+\beta+\gamma=9=3\beta\Rightarrow\beta\gamma\alpha=3$

Put $x=3\Rightarrow k=3^3-9(3)^2+26(3)=24$

(B) $\alpha\gamma=\beta^2\Rightarrow\alpha\beta\gamma=64=\beta^3\Rightarrow\beta=4$

Put $x=4\Rightarrow64-14(16)+4k-64=0\Rightarrow k=56$

(C) $\dfrac{2}{\beta}=\dfrac{1}{\alpha}+\dfrac{1}{\gamma}\Rightarrow\dfrac{1}{\alpha}+\dfrac{1}{\beta}+\dfrac{1}{\gamma}=\dfrac{1}{1/6}=\dfrac{3}{\beta}\Rightarrow\beta=\dfrac{1}{2}$

Put $x=\dfrac{1}{2}\Rightarrow\dfrac{6}{8}-\dfrac{k}{4}+\dfrac{6}{2}-1=0\Rightarrow k=11$

(D) $\alpha+\gamma=3\Rightarrow\alpha+\beta+\gamma=0=3+\beta\Rightarrow\beta=-3$

Put $x=-3\Rightarrow-27+3k+6=0\Rightarrow k=7$

87. $A=\dfrac{n(n+1)(2n+1)}{6}$

$B=\sum_{m=1}^{n}\dfrac{m(m+1)}{2}-\dfrac{1}{2}\dfrac{n(n+1)}{2}$

$\dfrac{1}{6}\sum_{m=1}^{n}(m+2)m(m+1)-m(m+1)(m-1)-\dfrac{n(n+1)}{4}$

$$=\frac{n(n+1)(n+2)}{6}-\frac{n(n+1)}{4}=\frac{n(n+1)(2n+1)}{12}$$

(B) $\dfrac{\sum a\left(b^2+c^2\right)}{abc}\geq\dfrac{a(2bc)+b(2ca)+c(2ab)}{abc}=6$

(C) $2\leq\sum\dfrac{x^2-1+1}{1-x}=2\sum-(x+1)+\dfrac{1}{1-x}=-8+2\sum\dfrac{1}{1-x}\geq-8+2\left(\dfrac{9}{2}\right)=1$

$$\left[\because\frac{3}{\sum\frac{1}{1-x}}\leq\frac{(1-x)}{3}=\frac{2}{3}\Rightarrow\sum\frac{1}{1-x}\geq\frac{9}{2}\right]$$

(D) $\because x,y,z\in N\Rightarrow\dfrac{1}{xy}\leq 1$

$\Rightarrow\dfrac{1}{xy}+\dfrac{1}{yz}+\dfrac{1}{2x}\leq 3$

$\therefore x=y=z=1$

88. (A) $xy^2z^2=16\Rightarrow\dfrac{x+\frac{y}{2}+\frac{y}{2}+\frac{z}{2}+\frac{z}{2}}{5}\geq\left(\dfrac{xy^2z^2}{16}\right)^{\frac{1}{5}}=1$

$\Rightarrow x+y+z\geq 5$

(B) $a+b+=3,\dfrac{3}{\frac{2}{a}+\frac{2}{a}+\frac{1}{b}}\leq\dfrac{\frac{a}{2}+\frac{a}{2}+b}{3}=1\Rightarrow\dfrac{4}{a}+\dfrac{1}{b}\geq 3$

(C) $a+c=2b\quad a+b>c\Rightarrow\dfrac{b}{c}>\dfrac{2}{3}$ and $b+c>a\Rightarrow\dfrac{b}{c}<2\Rightarrow\dfrac{2b}{c}\in\left(\dfrac{4}{3},4\right)$

(D) $3^{|\sin x|}\in[1,3],2^{-|\sec y|}\in\left(0,\dfrac{1}{2}\right],5\cos z\in[-5,5]$

$a=3^{|\sin x|}2^{-|\sec y|}+5\cos z$

$\Rightarrow a\in\left(-5,5+\dfrac{3}{2}\right],a\in\left(-5,\dfrac{13}{2}\right]$

89. (A) $\dfrac{\left(\frac{49a}{3}\right)3+\left(\frac{3b}{2}\right)2+c}{6}\geq\left(\left(\dfrac{49a}{3}\right)^3\left(\dfrac{3b}{2}\right)^2c\right)^{1/6}=\left(\dfrac{7^6a^3b^2c}{12}\right)^{1/6}$

$\Rightarrow 49a+3b+c\geq 6\times 7=42$

(B) $x=-t,t>0$

$2x^3-\dfrac{3}{x^2}=-\left(2t^3+\dfrac{3}{t^2}\right)\leq-5$

$$\because \frac{t^3+t^2+\frac{1}{t^2}+\frac{1}{t^2}+\frac{1}{t^2}}{5} \geq \left(\left(t^3\right)^2 \frac{1}{\left(t^2\right)^3}\right)^{1/5} = 1$$

$$\Rightarrow 2t^3+\frac{3}{t^2} \geq 5$$

(C) $$\left(\frac{\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{x^3}{5}+\frac{8-x^3}{3}+\frac{8-x^3}{3}+\frac{8-x^3}{3}}{8}\right)^8 \geq \left(\frac{8-x^3}{3}\right)\left(\frac{x^3}{5}\right)$$

$$\Rightarrow \left(8-x^3\right)x^5 \leq \left(3^3 5^5\right)^{1/3} = 3\times 5^{5/3}$$

(D) $$\frac{(x+x+x+x+x+x+x)+(y+y+y+y+y)}{12} \geq \left(x^7 y^5\right)^{1/12}$$

$$\Rightarrow (7x+5y) \geq 12a^{1/12}$$

$$\therefore 12a^{1/12} \geq 12$$

$$\Rightarrow a \geq 1$$

90. $b_1, b_2, b_3, b_4, b_5, b_6, b_7, b_8, b_9, b_{10}$

$= 1, r, r^2, \left(2r^2-r\right), (2r-1)^2, (2r-1)(3r-2), (3r-2)^2, (3r-2)$

$(4r-3), (4r-3)^2, (4r-3)(5r-4)$

$b_5+b_6=198$

$\Rightarrow (2r-1)(5r-3)=198$

$\Rightarrow r=5$

$r=\frac{39}{10}(rejected)$

$b_7=(3\times 5-2)^2=169$

$b_8=(3\times 5-2)(4\times 5-3)=221$

$b_9=(4\times 5-3)^2=289$

$b_{10}=(4\times 5-3)(4\times 5-4)=357$

91. $\log_2 5=2a, \log_5 2+\ln_5 3=b$

$$\Rightarrow \frac{1}{2a}+\log_5 3=b \Rightarrow \log_5 3=b-\frac{1}{2a}=\frac{2ab-1}{2a}$$

$$\log_3 2=\frac{\log_5 a}{\log_5 3}=\frac{\left(\frac{1}{2a}\right)}{\left(\frac{2ab-1}{2a}\right)}=\frac{1}{2ab-1}$$

92. $\ln 5(\ln 5+\ln(x+1))=\ln 2(\ln 2+\ln y)$

$\ln 2\ln(x+1)=\ln y\ln 5$

$\Rightarrow \ln^2 5+\ln(x+1)=\ln^2 2+\dfrac{\ln^2 2\ln(x+1)}{\ln 5}$

$\Rightarrow \ln^2 5-\ln^2 2=\ln(x+1)+\left(\dfrac{\ln^2 2-\ln^2 5}{\ln 5}\right)$

$\Rightarrow \ln(x+1)=-\ln 5$

$\Rightarrow x+1=\dfrac{1}{5}\Rightarrow x=-\dfrac{4}{5}$

93. Adding $\log_8(ab)+\log_4 a^2b^2=12$

$\Rightarrow \log_2(ab)+3\log_2(ab)=36$

$\Rightarrow (ab)^4=2^{36}$

$\Rightarrow ab=2^9=512$

94. $w=x^{24}\Rightarrow x=w^{1/24}$

$w=y^{40}\Rightarrow y=w^{1/40}$

$w=(xyz)^{12}\Rightarrow z=w^{\frac{1}{12}-\frac{1}{24}-\frac{1}{40}}=w^{\frac{1}{60}}$

$\Rightarrow \log_2 w=\log_{w^{1/60}} w=60$

95. $\log_2\left(\dfrac{1}{3}\log_2 x\right)=\dfrac{1}{3}\log_2(\log_2 x)$

$\Rightarrow \left(\dfrac{1}{3}\log_2 x\right)^3=\log_2 x$

$\Rightarrow (\log_2 x)^2=27$

$\Rightarrow \log_2 x=3\sqrt{3}$

96. The number $2\log_{(2000)^6}4+3\log_{(2000)^6}5$

$=\log_{(2000)^6}\left(4^2+5^3\right)=\dfrac{1}{6}$

97. $abc=b^3=6^6$

$\Rightarrow b=36$

Check $a=11,20,27,32,35$

$\Rightarrow a=27, b=36, c=48$

98. $(24\cos x)^2=(24\sin x)^3$

$\Rightarrow 24\sin^3 x+\sin^2 x-1=0$

$\Rightarrow (3\sin x-1)\left(8\sin^2 x+3\sin x+1\right)=0$

$\Rightarrow \sin x = \frac{1}{3}$

$\Rightarrow \cot^2 x = \cos ec^2 x - 1 = 9 - 1 = 8$

99. $n = b^2, 2b = n^2$

$\log_{b^2}(2b) = \log_n\left(n^2\right) = 2$

100. $\log_{10} x\left(3\log_{10} x - 2\right) = 1$

$\Rightarrow \log_{10} x = \frac{1}{3}\ or\ 1$

$\log_{10} x = -\frac{1}{3}$ is rejected

$\Rightarrow x = 10$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